SULIT

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION
SEMESTER II 2018/2019

CODE OF SUBJECT	**: SCSJ3323/ SCJ3323**
NAME OF SUBJECT	**: SOFTWARE ARCHITECTURE AND DESIGN**
YEAR / COURSE	**: 3/4/SCJ/4SCJ**
TIME	**: 3 Hours**
DATE	**:**
VENUE	**:**

ANSWER SCHEME:

THIS EXAMINATION BOOK CONSISTS OF TWO (2) PARTS:

- **PART A**: 5 STRUCTURED QUESTIONS (50 MARKS)
  Answer all the questions

- **PART B**: 2 PROBLEM SOLVING QUESTION (50 MARKS)
  Answer all the questions

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

Name	
I/C number	
Year/Course	
Section	
Lecturer	

PART A: STRUCTURED QUESTION **[60 Marks]**

Question 1 **[10 marks]**

As shown in **Figure 1**, you are assigned to evaluate the design of a security package. The Operator is a superclass which inherits its method *isAuthorized()* to the specific subclasses which are Administrator, Student, and Lecturer.

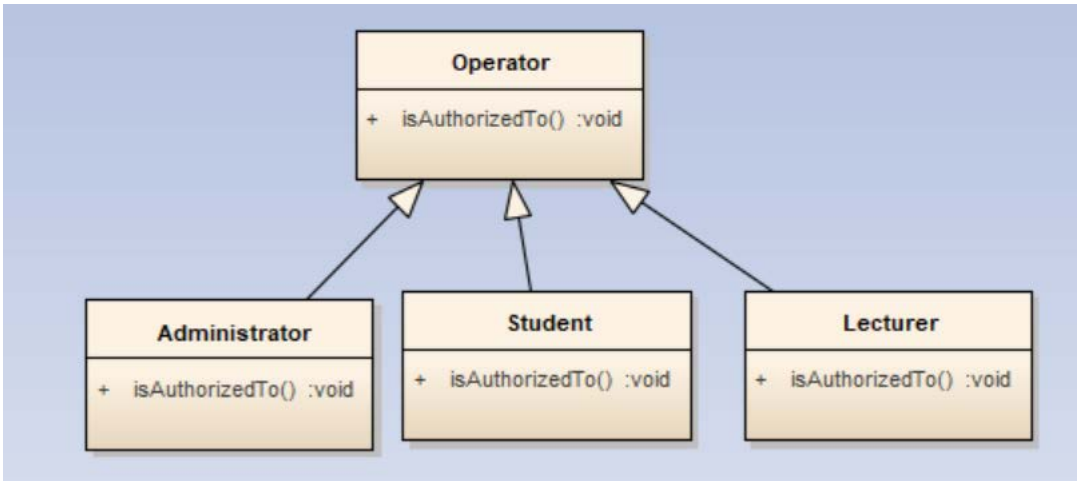
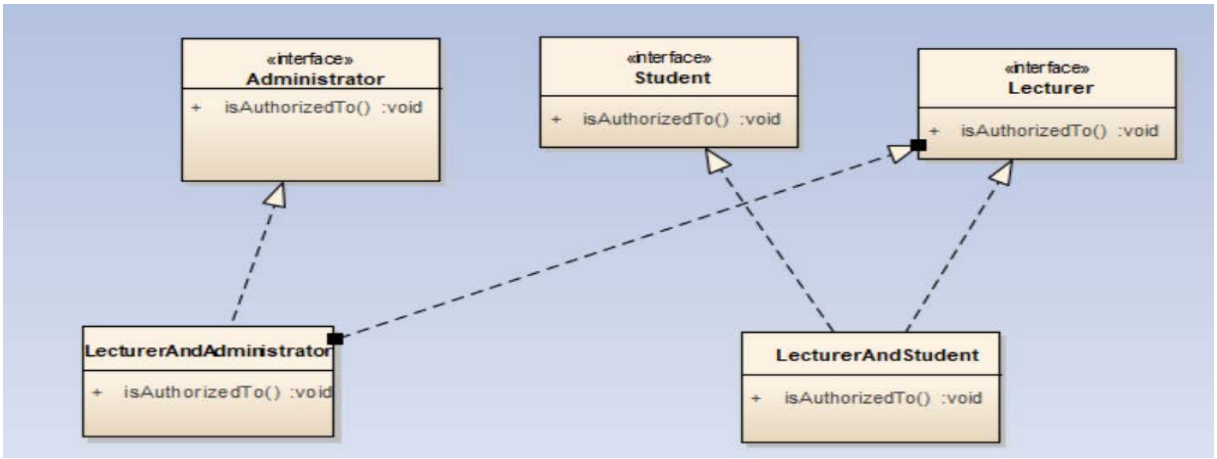


Figure 1

a. Analyse what would be the problem if a different individual can play different roles. For example, a lecturer can be an administrator. Therefore, LecturerAndAdministrator class has to be created. Or a lecturer can also be a student where LecturerAndStudent class has to be created. Describe the problem and give an example of the problem. [3 marks]
Problem – unmanageable class as the roles grows; when changing access right to Lecturer, the access right for LecturerAndAdministrator, LecturerAndStudent will have to be changed.

b. Identify what type of detailed design principle can be applied to this type of problem and justify why it is suitable. [2 marks]
Interface segregation principle where the each classes has its own interface. Any new class can realize suitable interfaces

c. Draw the class diagram to have a more manageable and maintainable and reusable clasess hierarchy based on the detailed design principle you have chosen in b.
[5 marks]



Question 2 **[10 marks]**

a. Briefly describe the main difference between Architectural Patterns and Design Patterns.

1. Architectural pattern is the reusable design solution (1)
2. Design patterns – **design pattern** is a general repeatable solution to a commonly occurring problem in software **design** at design details level (1m)

[2 marks].

b. Briefly describe the purpose of any FOUR (4) of the following design patterns. [4 marks].

i. Abstract Factory
Manage and encapsulate the creation of set of objects conceptually belong together that specifically represent specific family of pattern

ii. Observer
Behavioral pattern that standardizes the operations between objects that interoperate using a one-to-many relationship so that when one object changes state, all its dependents are notified and updated automatically.

iii. Adaptor
-Structural design pattern that deals with designing larger structures from existing classes or objects at run time
- integrate new design with existing ones via objects composition.

iv. Facade
-structural pattern that provide a simplified interface to complex subsystems
- give higher level of abstraction that liberate avoid/ set the clients free from responsibility of knowing the internal structure of various elements of the subsystems

c. Identify a suitable design pattern for each description below. [4 marks]

Description	**Design Pattern**
i. Which design pattern would you use in a situation where that encapsulates both the creational process and the representation of product objects.	i. Builder
ii. Which design pattern would you use to allow designers to compose (large) tree-like designs structures.	ii. Composite
iii. Which design pattern would you use to provide a standardized way for accessing and traversing objects in a collection data structure.	iii. Iterator
iv. Which design pattern would you use to prevent objects from being instantiated more than once in a running program.	iv. Singleton

Question 3 **[10 marks]**

a. The class diagram in **Figure 2** shows structure of a design pattern for a Cafe System that is composed with three subsystem; ColdPrep, HotPrep and Bar. Identify suitable design pattern that is represented by class diagram in **Figure 2**. Describe the roles of each class in that class diagram. [5 marks]

Client – Server Class
Façade – KitchenSection Class
Subsystem – ColdPrep, HotPrep, Bar Class

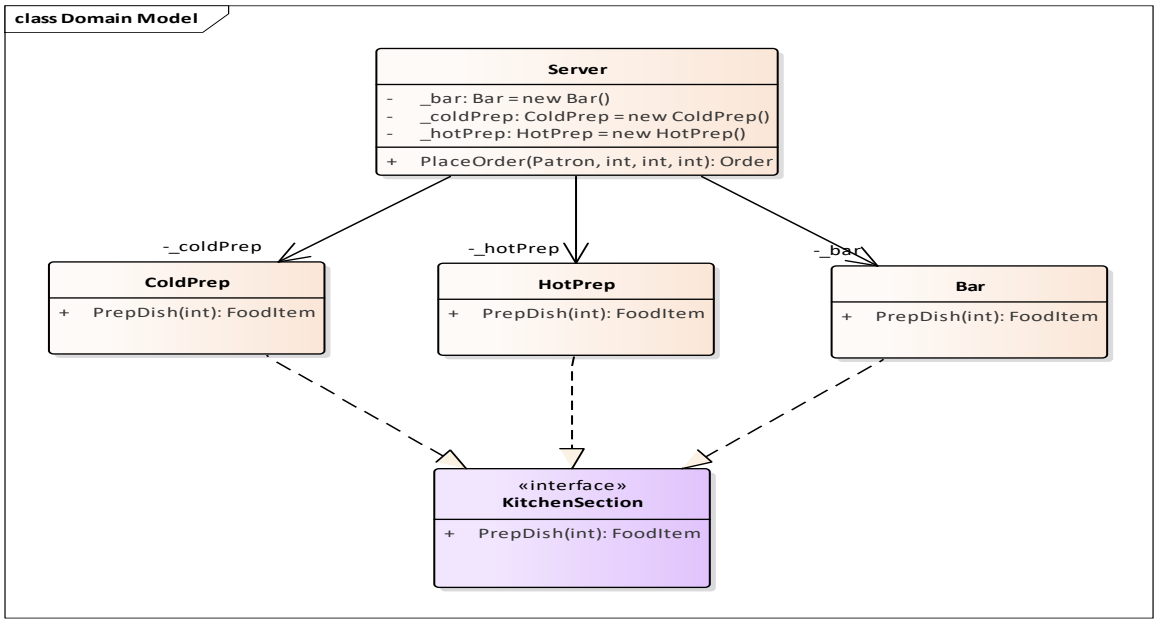


Figure 2

b. The diagram in **Figure 3** shows the information flow for generating different representation for Purchase Order form for the Procurement Subsystem. Which design pattern is suitable Figure 3 and draw the class diagram that applies your selected pattern, and describe TWO (2) justifications to support your design pattern selection. [5 marks]

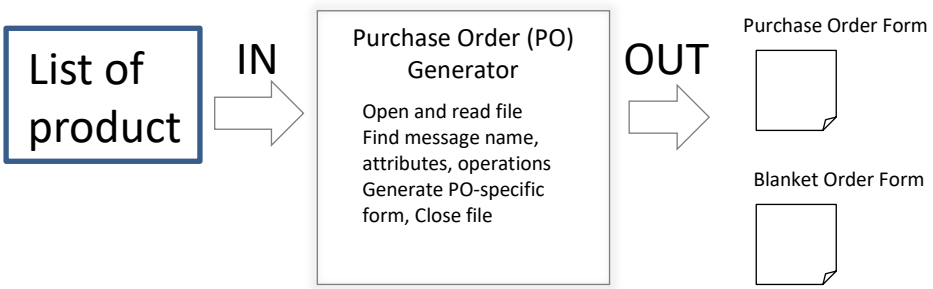
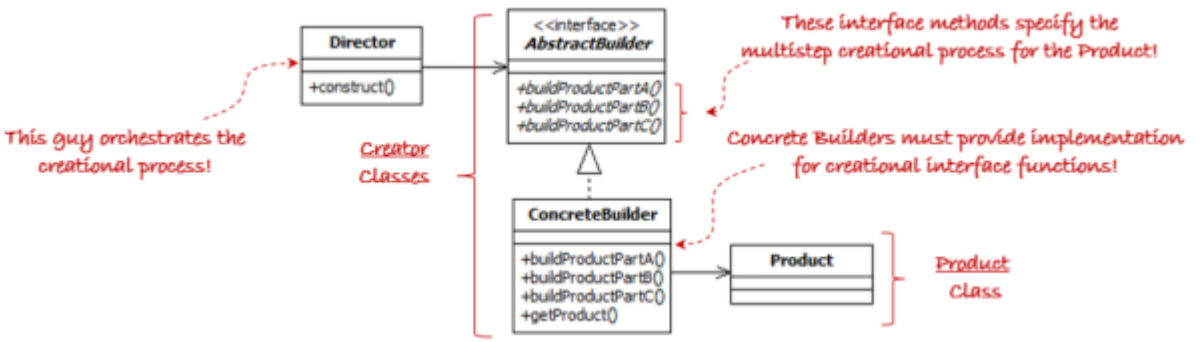


Figure 3

i. Builder design Pattern



TWO (2) justifications

Question 4

a. State the difference between state-based design and table based design as representations for behavioural construction design. [2 marks]

State-based design shows behaviour in the form of state while Table-based design shows behaviour in the form of decisions

b. **Table 1** shows a table-based design for SOFTWARE ORDERS. There are different conditions for giving a discount based on software orders. Discounts are given based on condition either the purchase is based on site license, educational discount or discount based on the number of copies purchased. Answer the following questions.

Table 2

Software Orders	Order1	Order2	Order3	Order4	Order5	Order6	Order7	Order8
SITE LICENSE	Y	Y	Y	Y	N	N	N	N
EDUCATIONAL DISCOUNT	Y	Y	N	N	Y	N	N	N
DISCOUNT FOR QUANTITY	Y	N	N	N	Y	Y	Y	N
Cost = Site License Cost	X	X	X	X				
Cost = Educational Cost X Copies purchase						X		
Cost = Discount Cost X Copies purchase							X	
Cost = Upgrade Cost X Copies purchase								X
Cost = (Educational Cost – Discount) X Copies purchase					X			

i. Calculate the number of distinct policies for software orders based on **Table 1**. [2 marks]

$2^{\text{no of condition}} = 2^3 = 8$ policies of software orders

If the students hav already reduced the policies into certain numbers (1 marks)

ii. Based on **Table 3**, there are several rules which are redundant. Identify THREE (3) decisions which can be optimized from the table and justify why it should be omitted. Write your answers based on **Table 3** format. [6 marks]

Table 3

Order No	Why it should be omitted?
6/7	Conditions are the same
8	Missing policies – there are basically no condition so there should be no action
1-4	Redundant actions – 2 or more order, results in identical actions

Question 5 **[10 marks]**

a. Define the TWO (2) main reasons why Software Design Management is important. [2 marks]

- Control/monitor design processes (1m)
- Manage the resources necessary for completing quality design artifacts according to organizational goals (1m)

b. Discuss the difference between Work Breakdown Structure (WBS) and Linear Responsibility Chart (LRC)? Describe a simple example of LRC that consists of 3 tasks and 2 staffs. [3 marks]

- WBS – The work breakdown structure (WBS) is a technique to represent, either graphically or in list format, a project modularized into task activities. (1m)
- LRC - plan for task assignments for staff (1m)
- LRC example (1m)

c. Various principles of ethics collectively ensure that professionals in the software engineering field adhere to high-levels of ethical conduct. Identify the suitable name of each software engineering principle of ethics according to the provided description.[5 marks]

SE Principle of Ethic	**Description**
i. Public	Software engineers shall act consistently with the public interest.

ii. _Product___	Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
iii. _Judgment__	Software engineers shall maintain integrity and independence in their professional judgment.
iv. _Management__	Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
v. _Profession__	Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

PART B: PROBLEM SOLVING **[50 Marks]**

Part B consists of **TWO (2)** essay questions. Answer **all the** questions. Each question carries 25 marks.

Question 1 **[25 marks]**

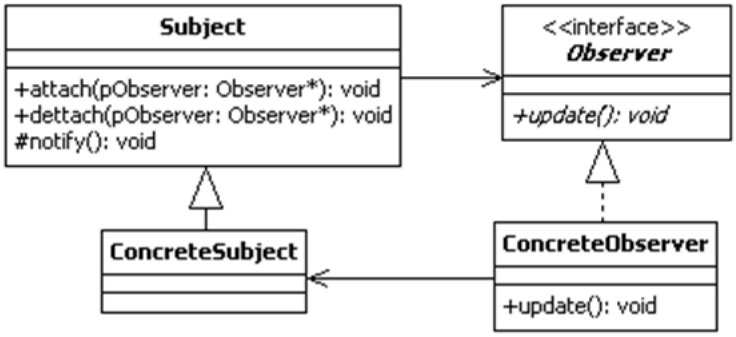
ABC bank is designing a system for fraud-alert notification that allows bankers, consumers and staff to receive notifications for anomaly money transaction (10 transaction per minute) via SMS, email and voice call. The system is based on anomaly transaction engine that interface with several fraud-alert data sources (debit card, credit card, eWallet, internet banking) where these data will be fused and analyzed and automatically decide whether the consumers are in danger (being fraud). The ABC banking company is interested to develop a decision engine which is responsible to send a notification of fraud to the customers via existing communication channel (SMS, email, voice). The design must be flexible so that other type of communication channels can be added in the near future.

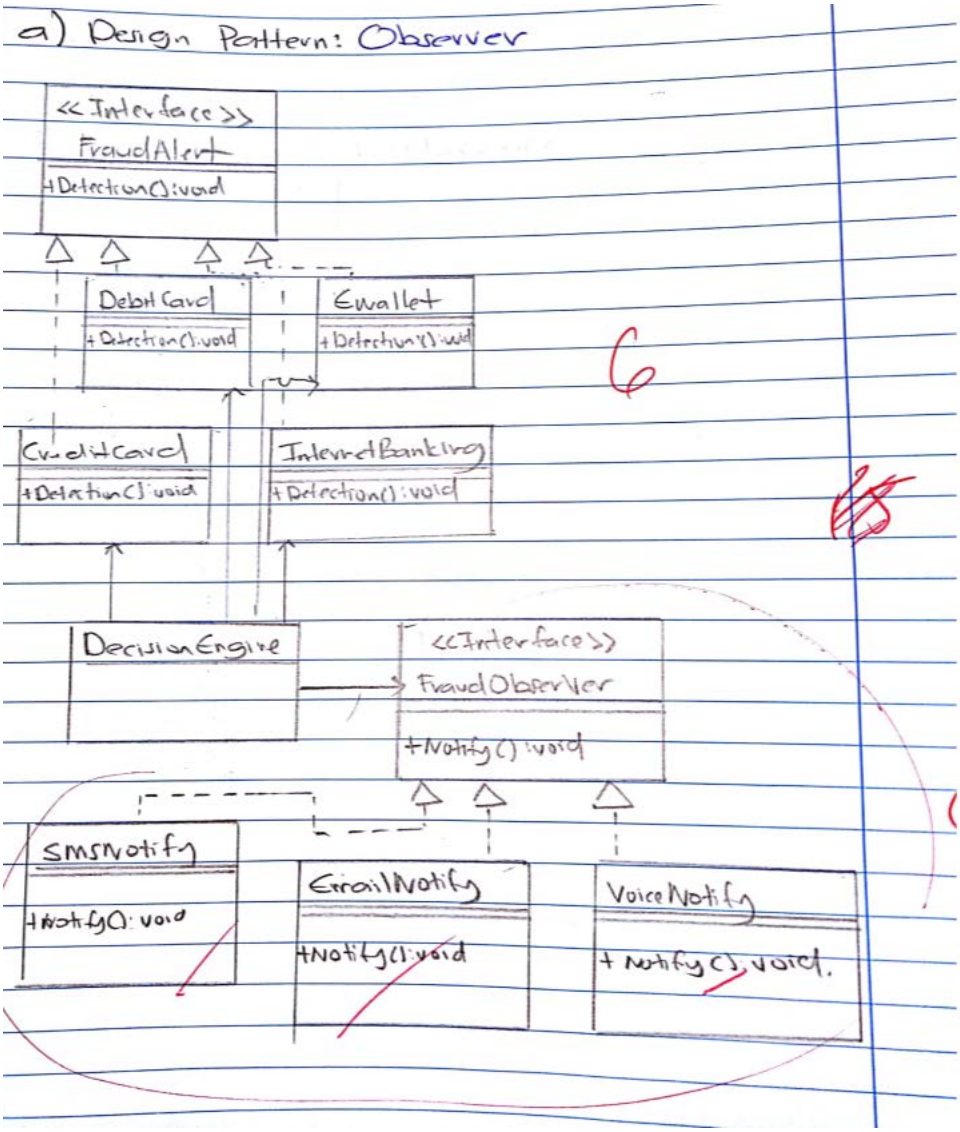
a. Draw a class diagram of the suitable Design Pattern and give TWO (2) justifications on why the design pattern is selected and describe TWO (2) benefits of using that design pattern.

2 benefits = 3 marks

Observer – Subject and Concrete Subject = 6 marks

- **Observer and Concrete Observer = 6 marks**





[15 marks]

b. Apply Adapter Design Pattern to the following design problem: [10 marks]

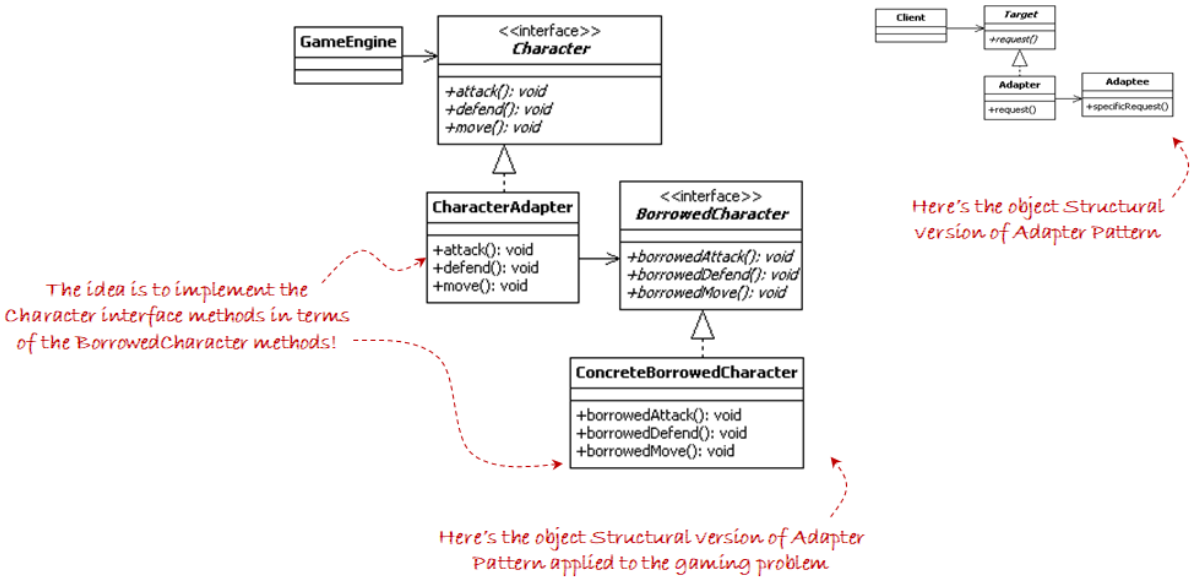
The XYZ company has developed a Spool Tracking System (STS), enable the workers at the ship yard to track a spool using RFID tag and mobile apps that being processed for production. STS can do the following function:

- *initiate()*
- *trackSpool()*
- *registerSpoolID()*
- *scanSpool()*
- *end()*

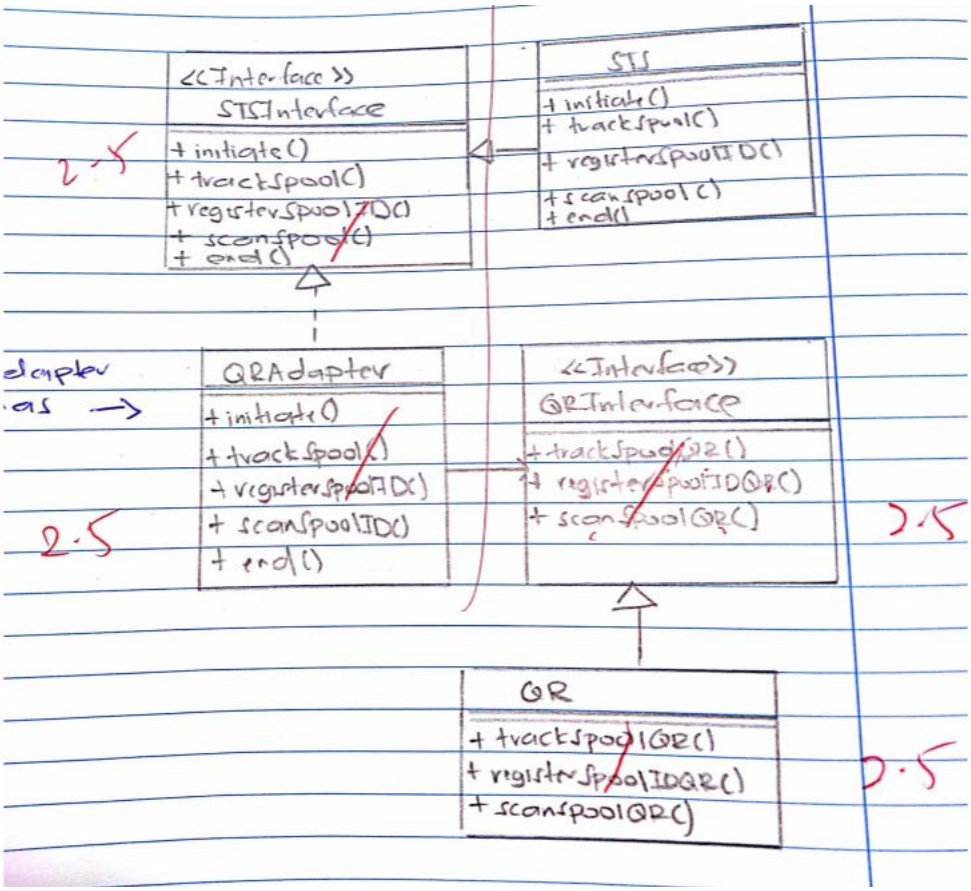
As an appointed consultant, you have to improve the STS class so that the *trackSpool(), registerSpoolID* and *scanSpool()* may perform using different tag (QR Code). However, it is a time taking task to modify the code and luckily, you manage to find the code with the third party company for QR code mechanism. The third party code consists of the following operations; *trackSpoolQR(), registerSpoolIDQR()* and *scanSpoolQR()*. Thus you want to reuse the class diagram and code to improve the existing STS. For this purpose, it is suitable to apply Adapter Design Pattern.

Your Task: Draw **a class diagram** so that STS can use the functions from the third party code (QR). Highlight the adapter class in your diagram.

ADAPTER DESIGN PATTERN EXAMPLE



Each adapter and adaptee classes = 2.5 marks



Question 2 **[25 marks]**

a. Describe TWO (2) importance of having a construction style. [2 marks]

- ensure high quality and consistent implementation of programming problem
- ensure consistency, simplicity and clarity of code

b. Discuss how Programming Design Language (PDL) can help in realizing the construction style. [2 marks]

An approach which the flow of the program is written without focusing on the target programming language.

-This will help those who do not have any background knowledge on programming language. PDL will also help in the complete documentation of the program.

c. Based on the codes given for calculating tax in **Figure 4**, analyse and distinguished THREE (3) different violations of software construction styles and describe each styles which are violated. [6 marks]

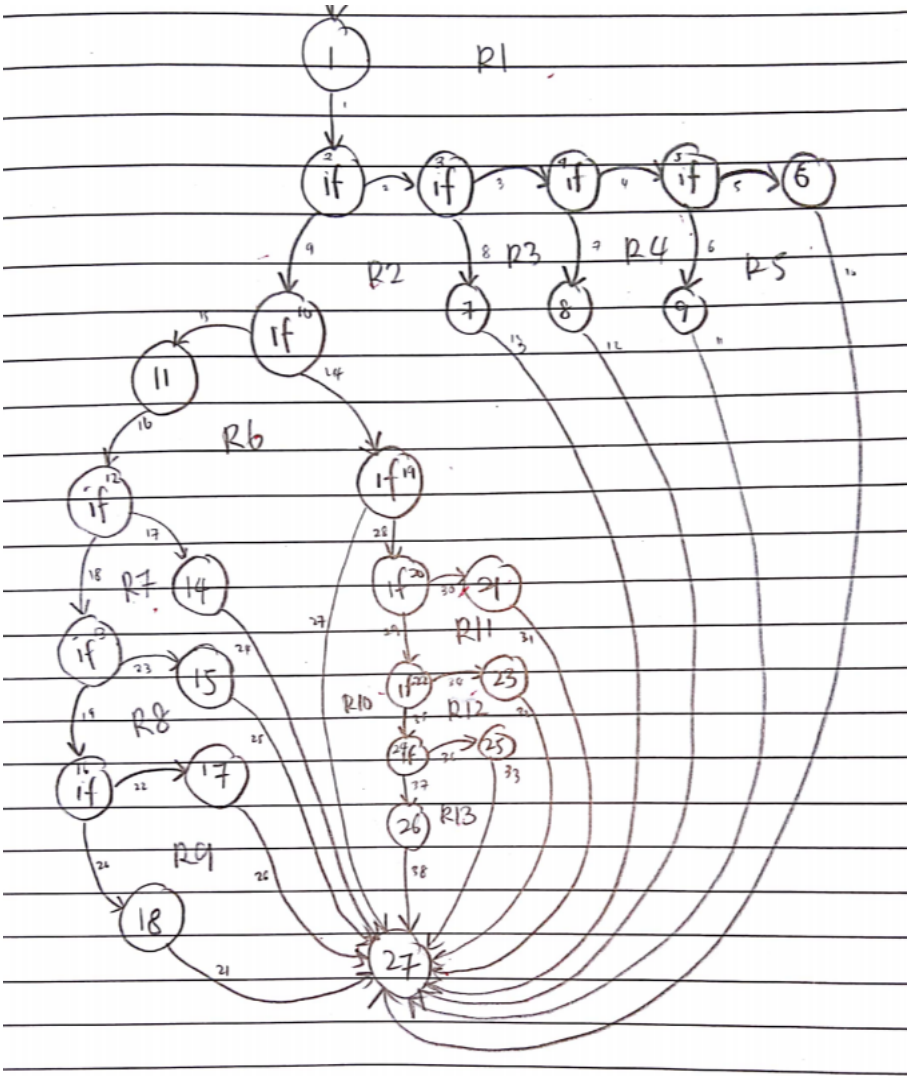
Formatting conventions – do not use white spaces, indentation and bracket placement
Naming convention – the name of the variable do not follow a meaningful convention
Documentation convention – the code do not have any comments

```
GetƒTaxƒRatesƒbasedƒonƒNumberƒDependents,ƒPayrate
CalculateƒIncomeƒTaxƒ= PA *TR
If D < Min Then
CalculateA =PeriodBasePay*FICAER
Calculate CorpFICA = PeriodBasePay*FICACorpRate EndIf
If StateTaxRequired Then Get
StateTaxRatebasedonState,NumberDependents,Payrate Calculate
StateTax=PeriodPayAmount*ƒStateTaxRate EndIf
If OvertimePay >0 Then
CalculateOvertimeIncomeTax=PeriodOvertimePay*IncomeTaxRate
AddOvertimeIncomeTaxtoIncometax If YTDPay<FICAMaximumAmount Then
CalculateEmpOvertimeFICA =PeriodOvertimePayFICAEmployeeRate
CalculateCorpOvertimeFICA=PeriodOvertimePay*FICACorpRate EndIf
If StateTaxRequired Then
CalculateƒStateOvertimeTax =PeriodOvertimePay*StateTaxRate
EndIf
EndIf
```

Figure 4

d. Figure 5 shows the code for calculating Phone Discount Function. The following questions are based on Figure 5.

i. Convert the code into flow based design. [5 marks]



ii. Using the right equation, calculate cyclomatic complexity based on the graph, vertices edges, and components. [3 marks]

V(g) = e – n + 2p

= 38-27+2(1)

= 13

iii. Using the right equation, calculate cyclomatic complexity based on programs decision construct. [3 marks]

***c* = *π* + 1**

***π* = no of if = 12 c= 12+1 = 13**

iv. Using the right equation, calculate cyclomatic complexity based on the program's control graph.

2 = n – e + r

2 = 27-38+r

2 = -11+r

r =13

[3 marks]

v. Apply the threshold value for the cyclomatic complexity and identify whether the quality of the code is acceptable. [1 marks]

Threshold value = 10

13>10 so the quality of code is not acceptable

***** any value which is not accurate ie cyclomatic complexity = 12 just deduct marks at the flow graph and not at the calculation ii-iv**

- **Provided the formula and the calculation is correct based on the lecture notes**

Figure 5